

CATANIA FONTANAROSSA, Italy

WMO: 164600

Lat: 37.467N

Lon: 15.066E

Elev: 39

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.5	37.1	26.2	20.2	45.9	28.8	22.8	45.0	26.6	55.1	23.5	55.7	6.4	240	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.2	94.5	73.1	91.2	73.0	88.4	72.6	79.0	86.0	77.3	85.1	75.8	84.1	10.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	141.5	82.5	75.2	132.2	81.5	73.4	124.2	80.5	42.5	85.3	40.7	84.6	39.2	84.0	89.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.1	18.9	16.6	DB	30.4	103.4	1.7	5.0	29.2	107.0	28.2	109.9	27.2	112.7	26.0	116.3
			WB	29.0	80.8	1.8	3.5	27.7	83.3	26.6	85.3	25.6	87.2	24.2	89.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.7	50.7	51.0	54.1	58.5	65.3	73.0	78.0	78.9	73.9	67.2	59.5	53.2	(6)
	DBStd	10.92	4.41	4.34	4.49	3.89	4.34	4.62	3.61	3.49	4.10	4.17	4.71	4.52	(7)
	HDD50	112	44	35	11	1	0	0	0	0	0	0	2	19	(8)
	HDD65	1984	444	391	339	196	48	1	0	0	1	25	173	366	(9)
	CDD50	5105	65	64	139	257	475	689	867	895	717	533	286	118	(10)
	CDD65	1501	0	0	1	2	57	240	402	430	268	94	7	0	(11)
	CDH74	14546	0	1	21	26	421	2263	4559	4753	2062	412	26	2	(12)
	CDH80	5426	0	0	7	4	81	735	1938	2037	572	51	1	0	(13)
(14) Wind	WSAvg	7.6	7.6	8.2	8.3	8.2	7.8	7.5	7.5	7.3	7.4	6.9	7.1	7.5	(14)
(15) Precipitation	PrecAvg	22.2	3.0	2.1	1.9	1.4	0.7	0.2	0.2	0.4	1.6	5.0	2.4	3.4	(15)
	PrecMax	36.5	10.2	5.8	6.2	6.1	4.5	0.7	2.1	2.1	8.2	14.6	9.0	15.9	(16)
	PrecMin	8.7	0.4	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.1	0.0	(17)
	PrecStd	7.9	2.6	1.5	1.4	1.3	1.0	0.3	0.5	0.5	1.7	3.9	2.4	3.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.0	69.4	75.4	77.1	87.3	96.4	99.9	99.6	93.4	85.1	77.1	70.1	(19)
		MCWB	56.9	56.2	58.9	60.4	65.9	69.7	74.1	74.4	70.6	68.4	64.9	60.2	(20)
	2%	DB	64.4	65.8	69.6	73.1	81.0	89.8	93.5	94.1	88.0	80.7	73.5	66.4	(21)
		MCWB	54.4	54.6	56.9	59.3	65.3	70.7	73.3	74.0	71.7	67.8	63.6	57.4	(22)
	5%	DB	62.1	62.7	66.3	70.0	78.5	86.4	90.8	91.1	85.6	78.5	71.2	64.3	(23)
		MCWB	53.9	53.8	55.5	58.6	64.5	70.4	72.8	73.9	70.9	67.1	62.8	56.6	(24)
	10%	DB	59.4	60.6	64.0	68.0	75.5	84.1	88.1	88.2	82.8	75.7	68.3	62.3	(25)
		MCWB	52.6	53.0	54.9	58.1	63.4	69.2	72.2	73.6	70.4	66.5	61.1	55.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.0	59.2	60.9	64.2	70.2	77.8	80.5	82.0	78.9	74.4	69.4	62.5	(27)
		MCDB	63.2	63.8	68.7	69.5	77.4	85.4	88.8	87.2	83.6	77.4	72.4	66.9	(28)
	2%	WB	57.5	57.2	59.0	62.4	68.2	75.0	78.1	79.6	76.3	71.8	66.6	60.3	(29)
		MCDB	61.9	62.0	65.0	68.5	76.1	83.1	86.0	86.2	81.9	76.5	70.8	64.2	(30)
	5%	WB	55.5	55.8	57.6	60.9	66.7	73.3	76.4	78.1	74.5	69.9	64.4	58.7	(31)
		MCDB	60.4	60.6	63.3	66.6	74.7	82.2	85.0	85.7	80.7	75.2	68.8	62.8	(32)
	10%	WB	53.7	54.1	56.4	59.7	65.2	71.5	75.0	76.5	72.9	68.4	62.5	56.5	(33)
		MCDB	58.6	59.6	62.1	65.2	73.1	81.0	84.2	85.1	79.5	74.2	67.1	61.0	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.0	18.8	19.3	19.0	19.9	20.5	21.1	20.2	18.5	17.5	17.6	17.2	(35)
		MCDBR	21.1	21.7	23.3	22.8	23.9	23.9	24.6	23.6	21.7	20.3	20.4	19.8	(36)
	5% WB	MCWBR	13.6	13.5	13.8	12.6	11.3	11.4	12.7	12.8	10.9	11.2	12.7	12.7	(37)
		MCWBR	17.6	18.9	20.0	18.3	19.9	21.3	21.1	21.0	18.7	17.4	17.4	16.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.352	0.376	0.424	0.461	0.463	0.447	0.447	0.455	0.454	0.427	0.388	0.358	(40)
	taud		2.414	2.325	2.180	2.100	2.142	2.227	2.246	2.222	2.229	2.269	2.343	2.408	(41)
	Ebn at Noon		261	268	265	261	262	266	265	260	252	247	244	250	(42)
	Edh at Noon		28	34	43	50	49	45	43	43	41	36	30	26	(43)
(44) All-Sky Solar Radiation	RadAvg		725	972	1370	1742	2082	2331	2400	2108	1567	1137	796	643	(44)
	RadStd		59	89	104	121	115	107	75	116	85	85	61	56	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	-2.92	-3.37	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	-0.77	-1.10	N/A	N/A	N/A	N/A

Nomenclature: See separate page